

## Lecture 8

Microeconomic Theory III – ECON 709  
Fall 2026

## Recap' lecture 7

Consumer preferences:

- ▶  $K$  commodities
- ▶ a *bundle*  $x = (x_1, \dots, x_K)$  is a combination of the  $K$  commodities where  $x_k$  is the quantity of commodity  $k$
- ▶ the set of all available alternatives,  $X$ , is the *consumption set*
- ▶ assume  $X = \mathbb{R}_+^K = \{x = (x_1, \dots, x_K) \mid \text{for all } k, x_k \geq 0\}$
- ▶ from now on, assume that a consumer's preference relation  $\succsim$  over  $X$  is rational
- ▶ A preference relation  $\succsim$  on  $X$  is **monotone** if for all  $x, y \in X$ ,  $x_k \geq y_k$  for all  $k$  implies  $x \succsim y$ , and  $x_k > y_k$  for all  $k$  implies  $x \succ y$ .
- ▶ A preference relation  $\succsim$  on  $X$  is **strongly monotone** if for all  $x, y \in X$ ,  $x_k \geq y_k$  for all  $k$  and  $x \neq y$  implies  $x \succ y$ .
- ▶ Our definition of monotonicity is a stronger version of MWG.

# Monotonicity and Utility Representations

- ▶ A function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is strictly increasing if  $x_k \geq y_k$  for all  $k$  implies  $f(x) \geq f(y)$ , and if  $x_k > y_k$  for all  $k$  implies  $f(x) > f(y)$ .

A function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is strongly increasing if  $x_k \geq y_k$  for all  $k$  and  $x \neq y$  implies  $f(x) > f(y)$ .

The following result is easy to check:

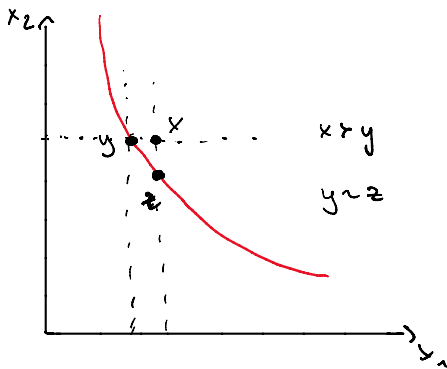
- ▶ If  $u$  represents preference relation  $\succsim$ ,
  - ▶  $\succsim$  is monotone if and only if  $u$  is strictly increasing.
  - ▶  $\succsim$  is strongly monotone if and only if  $u$  is strongly increasing.

## Examples of (strongly) monotone preference relations

- ▶ If a preference relation  $\succsim$  on  $X = \mathbb{R}_+^2$  is represented by  $u(x) = \min \{x_1, x_2\}$ , then  $\succsim$  is monotone, but not strongly monotonic.
- ▶ If a preference relation  $\succsim$  on  $X = \mathbb{R}_+^2$  is represented by  $u(x) = x_1 + x_2$ , then  $\succsim$  strongly monotone.

## Strong monotonicity implies strictly decreasing indifference curves

Let  $K = 2$ . If  $y \sim z$  and  $y_1 < z_1$ , then  $z_2 < y_2$  by strong monotonicity. If  $z_2 \geq y_2$ , then strong monotonicity would imply that  $z \succ y$ , a contradiction.



# Local Non-Satiation

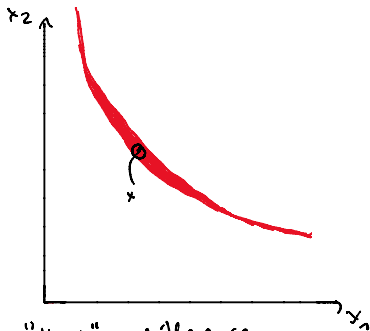
A weaker "more is better" property than monotonicity:

## Definition 13

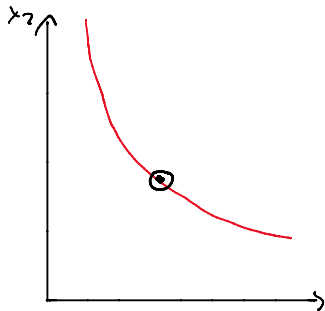
The preference relation  $\succsim$  is *locally non-satiated* if for every  $x \in X$  and every  $\epsilon > 0$ , there exists  $y \in X$  such that  $\|y - x\| \leq \epsilon$  and  $y \succ x$ .

$\|x - y\|$  is the Euclidean distance between points  $x$  and  $y$ .

- ▶ Local non-satiation says that for any  $x$  and any arbitrarily small distance away from  $x$ , denoted by  $\epsilon > 0$ , there is another bundle  $y$  within this distance from  $x$  that is strictly preferred to  $x$ .
- ▶ A preference relation with "thick" indifference sets is not locally non-satiated.



"thick" indifference curves violate local non-satiability.



local non-satiability  $x_1$  implies "thin" indifference curves

## Relation between Local Non-Satiation and Monotonicity

- ▶ If a preference relation is monotonic, then it is also non-satiated:

For any  $x$  and any  $\epsilon > 0$ ,  $y$  with  $y_k \in (x_k, x_k + \epsilon)$  for all  $k$  is strictly preferred to  $x$  by monotonicity.

- ▶ But not every locally non-satiated preference relation is monotonic:

Suppose  $X = \mathbb{R}_+^2$  and let  $x \succsim y$  if and only if  $|x_1 - x_2| \geq |y_1 - y_2|$ . This is a preference relation which is locally non-satiated and not monotonic.

# Convexity: "Moderation is at least as good as the worse of two extremes"

Convex preferences capture two related intuitions in situations where we can speak of an "in between" two alternatives:

- ▶ If  $x$  is weakly preferred over  $y$ , then any move from  $y$  to  $x$  is also a weak improvement over  $y$ :  
Suppose you prefer living in the center of Montreal over living in the suburbs, then any move closer to the center of Montreal is better than living in the suburb.
- ▶ If  $z$  is between  $x$  and  $y$ , then it is impossible that both  $x$  and  $y$  are better than  $z$ :  
Consider preferences over political candidates. Any candidate between the most left and the most right is weakly preferred over the worse of the two extreme candidates.

Does convexity always make sense? Consider convex combinations of milk and orange juice...

# Definition of Convexity

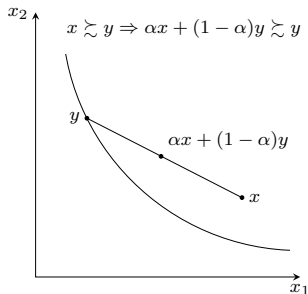
We will see now 3 different definitions of convexity each of which reveals a different intuition but which are all equivalent.

Thus, if we say a preference relation is convex, you can have any of the following three definitions in mind.

### 3 Equivalent Convexity Definitions

#### Definition 14 (Convexity 1.)

The preference relation  $\succsim$  satisfies convexity 1, if  $x \succsim y$  implies  $\alpha x + (1 - \alpha)y \succsim y$  for all  $\alpha \in [0, 1]$ .



Convexity 1

If  $x$  is weakly preferred over  $y$ , then any move from  $y$  to  $x$  is also a weak improvement over  $y$ .

### Definition 15 (Convexity 2.)

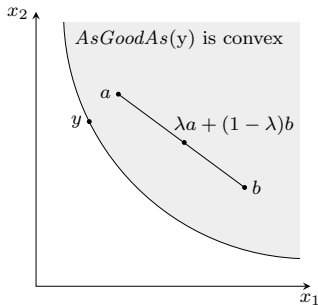
*The preference relation  $\succsim$  satisfies convexity 2, if for all  $x, y$  and  $z = \alpha x + (1 - \alpha)y$  with  $\alpha \in [0, 1]$ ,  $z \succsim x$  or  $z \succsim y$ .*

If  $z$  is between  $x$  and  $y$ , then it is impossible that both  $x$  and  $y$  are better than  $z$ ;

$z$  is at least as good as the worse alternative among  $x$  and  $y$ .

## Definition 16 (Convexity 3.)

The preference relation  $\succsim$  satisfies convexity 3, if for all  $x \in X$ , the upper contour set  $\{y | y \succsim x\}$  is convex; that is, if  $y \succsim x$  and  $z \succsim x$ , then  $\alpha y + (1 - \alpha)z \succsim x$  for any  $\alpha \in [0, 1]$ .



Convexity 3

C3 captures the intuition that if both  $z_1$  and  $z_2$  are better than  $y$ , then an average of  $z_1$  and  $z_2$  is also preferred to  $y$ .

# Equivalence Proof

## Proposition 12

*If the preference relation  $\succsim$  satisfies one of the conditions convexity 1, convexity 2, and convexity 3, then it satisfies the other two as well.*

### Proof.

$C1 \Rightarrow C2$ : Assume that  $\succsim$  satisfies convexity 1 and let  $x, y, z \in X$  such that  $z = \alpha x + (1 - \alpha)y$  for some  $\alpha \in (0, 1)$ . Without loss of generality, assume that  $x \succsim y$ . By convexity 1, we have  $z \succsim y$ . Thus,  $\succsim$  satisfies convexity 2.

$C2 \Rightarrow C3$ : Assume that  $\succsim$  satisfies convexity 2 and let  $y \succsim x$  and  $z \succsim x$ . Then, by convexity 2,  $\alpha y + (1 - \alpha)z$  is at least as good as either  $y$  or  $z$  (or both). In any case, by transitivity,  $\alpha y + (1 - \alpha)z \succsim x$ , and thus  $\succsim$  satisfies convexity 3.

$C3 \Rightarrow C1$ : Assume that  $\succsim$  satisfies convexity 3. If  $x \succsim y$ , then both  $x \succsim y$  and  $y \succsim y$  and thus  $\alpha x + (1 - \alpha)y \succsim y$ . Thus,  $\succsim$  satisfies convexity 1.

# Strict Convexity

Stronger version of convexity:

"An average is **strictly** preferred to the worse one of the two extremes."

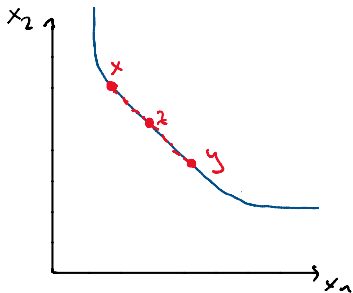
## Definition 17

*The preference relation  $\succsim$  is **strictly convex**, if  $y \succsim x$ ,  $z \succsim x$ , and  $y \neq z$  imply  $\alpha y + (1 - \alpha)z \succ x$  for all  $\alpha \in (0, 1)$ .*

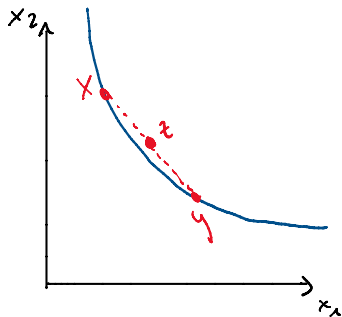
Above is the strict convexity analogue of C3.

Strict convexity analogue of definitions C1 and C2: strict preference implication.

Still equivalence between strict convexity analogues of definitions C1-C3.



- $x \succeq y$ ,  $z \succeq y$  and not  $z \succeq x$ .
- convex, but not strictly convex prefs.



- $x \succeq y$  and  $z \succ y$ ,
- strictly convex prefs.

## Example of Convex Preferences

Let  $X = \mathbb{R}_+^2$  and the preference relation  $\succsim$  be defined by  $x \succsim y$  if and only if  $x_1 + x_2 \geq y_1 + y_2$ .

- ▶  $\succsim$  is convex:

Let  $x \succsim y$ , i.e.  $x_1 + x_2 \geq y_1 + y_2$ . Then

$\alpha x + (1 - \alpha)y = (\alpha x_1 + (1 - \alpha)y_1, \alpha x_2 + (1 - \alpha)y_2)$  for  $\alpha \in (0, 1)$ . Now

$$\begin{aligned} & [\alpha x_1 + (1 - \alpha)y_1] + [\alpha x_2 + (1 - \alpha)y_2] \\ &= \alpha(x_1 + x_2) + (1 - \alpha)[y_1 + y_2] \\ &\geq \alpha(y_1 + y_2) + (1 - \alpha)[y_1 + y_2] = y_1 + y_2. \end{aligned}$$

Thus  $\alpha x + (1 - \alpha)y \succsim y$ .

- ▶  $\succsim$  is not strictly convex:

Consider bundles  $(1, 0)$  and  $(0, 1)$ . We know that

$(1, 0) \succsim (0, 1)$  and  $(0, 1) \succsim (0, 1)$ , and  $(1, 0) \neq (0, 1)$ . Take  $\frac{1}{2}(1, 0) + \frac{1}{2}(0, 1) = (\frac{1}{2}, \frac{1}{2})$ . However  $(\frac{1}{2}, \frac{1}{2}) \sim (0, 1)$  and not  $(\frac{1}{2}, \frac{1}{2}) \succ (0, 1)$ .

# Utility Representations of Convex Preferences

- ▶ A function  $u : X \rightarrow \mathbb{R}$  is concave if for all  $x, y$ , and  $\lambda \in [0, 1]$  we have  $u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y)$ , and is quasi-concave if for all  $y$ , the set  $\{x | u(x) \geq u(y)\}$  is convex, or equivalently, if  $u(\lambda x + (1 - \lambda)y) \geq \min \{u(x), u(y)\}$  for all  $x, y$  and  $\lambda \in [0, 1]$ .
- ▶ Trivially, any concave function is also quasi-concave.

## Proposition 13

*If a preference relation  $\succsim$  is represented by a utility function, then  $\succsim$  is convex if and only if  $u$  is quasi-concave.*

Proof left as exercise.

Thus, a concave utility function implies convex preferences.

However, convexity of a preference relation does not imply that its utility representation is concave!

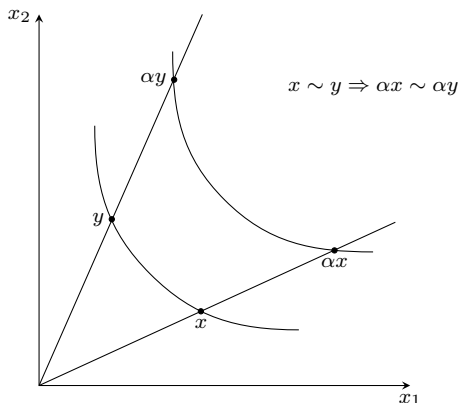
# Special Classes of Consumer Preferences

For some classes of preferences, it is possible to deduce the entire preference relation from a single indifference set, for example, homothetic or quasi-linear preferences...

# Homothetic Preferences

## Definition 18

A monotone preference relation  $\succsim$  is **homothetic** if for all  $x, y$ ,  $x \sim y$  implies that  $\alpha x \sim \alpha y$  for any  $\alpha \geq 0$ .



# Utility Representation of Homothetic Preferences

- ▶ For example, preferences represented by a “Cobb-Douglas” utility function  $\prod_{k=1,\dots,K} x_k^{\beta_k}$  with  $\beta_k > 0$  are homothetic.
- ▶ More generally, any preference relation represented by a utility function  $u$  that is homogeneous of any degree  $\lambda$  (that is  $u(\alpha x) = \alpha^\lambda u(x)$ ) is homothetic:

$$\begin{aligned} x \sim y &\Leftrightarrow u(x) = u(y) \Leftrightarrow \alpha^\lambda u(x) = \alpha^\lambda u(y) \\ &\Leftrightarrow u(\alpha x) = u(\alpha y) \Leftrightarrow \alpha x \sim \alpha y. \end{aligned}$$

## Proposition 14

*A monotone preference relation  $\succsim$  on  $X = \mathbb{R}_+^K$  is continuous and homothetic if and only if  $\succsim$  can be represented by a continuous utility function that is homogeneous of degree one.*

### Proof.

$\Rightarrow$ : We already proved for Prop. 8 that for any bundle  $x$  there exists a unique bundle  $(t(x), \dots, t(x))$  on the main diagonal, so that  $x \sim (t(x), \dots, t(x))$ , and that the function  $u(x) = t(x)$  is a continuous utility representation of  $\succsim$ . By the assumption that the preferences are homothetic,  $\alpha x \sim (\alpha t(x), \dots, \alpha t(x))$  and thus  $u(\alpha x) = \alpha t(x) = \alpha u(x)$ , that is  $u$  is homogeneous of degree 1.

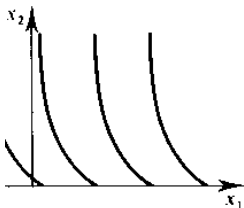
$\Leftarrow$ : By our previous results, since  $u$  is continuous and homogenous of degree  $\lambda$ ,  $\succsim$  is continuous and homothetic. □

# Quasi-linear Preferences

## Definition 19

A preference relation  $\succsim$  is **quasi-linear** in commodity 1 (referred to as "numeraire"), if

1. All the indifference sets are parallel displacements of each other along the axis of commodity 1. That is, if  $x \sim y$ , then  $(x + \alpha e_1) \sim (y + \alpha e_1)$  where  $e_1 = (1, 0, \dots, 0)$  and any  $\alpha \in \mathbb{R}$ .
  2. Good 1 is desirable; that is  $x + \alpha e_1 \succ x$  for all  $x$  and  $\alpha > 0$ .
- If  $I$  is an indifference curve, then the set  $I_\alpha = \{x \mid \text{there exists } y \in I \text{ such that } x = y + (\alpha, 0, \dots, 0)\}$  is an indifference curve as well.



## The numeraire as “money”

- ▶ The most common interpretation of commodity 1 is “money”, left over to buy other commodities not included in the commodities  $2, \dots, K$ .
- ▶ Part 1 of the definition implies that adding an equal amount of money to two indifferent bundles does not change the trade off in terms of preferences between the commodities (indifference remains); in other words, there are no “wealth effects”.
- ▶ Part 2 of the definition implies that money is desirable.
- ▶ Quasi-linearity in commodity 1, continuity and monotonicity of  $\succsim$  implies that for every  $(x_2, \dots, x_K)$  there is a number  $v(x_2, \dots, x_K)$  such that  $(0, x_2, \dots, x_K) \sim (v(x_2, \dots, x_K), 0, \dots, 0)$ . (For a proof refer to Rubinstein.)

That is, for every commodity bundle  $x$  with zero money, there exists an amount of money  $m$  such that there is indifference between having  $x$  and having  $m$ .

# Utility Representation of Quasi-Linear Preferences

- ▶ If a preference relation  $\succsim$  is represented by a utility function  $u(x) = x_1 + v(x_2, \dots, x_K)$  for some function  $v : \mathbb{R}_+^{K-1} \rightarrow \mathbb{R}$ , then  $\succsim$  is quasi-linear in commodity 1:
  - ▶ First,  $x \sim y \Leftrightarrow u(x) = u(y) \Leftrightarrow u(x) + \alpha = u(y) + \alpha \Leftrightarrow (x_1 + \alpha) + v(x_2, \dots, x_K) = (y_1 + \alpha) + v(y_2, \dots, y_K) \Leftrightarrow u(x + \alpha e_1) = u(y + \alpha e_1) \Leftrightarrow x + \alpha e_1 \sim y + \alpha e_1$ .
  - ▶ Second, consider  $x + \alpha e_1$  and  $x$  where  $\alpha > 0$ . Then  $u(x + \alpha e_1) = x_1 + \alpha + v(x_2, \dots, x_K) > u(x) = x_1 + v(x_2, \dots, x_K)$ . Thus  $x + \alpha e_1 \succ x$ .
- ▶ Intuition for the utility function  $u(x) = x_1 + v(x_2, \dots, x_K)$ :
  - ▶ The utility from bundle  $x$  is some “sub-utility” from  $(x_2, \dots, x_K)$  plus the amount of money left over.
  - ▶ If  $x \succ y$ , then  $u(x) - u(y) = m \Leftrightarrow u(x) = m + u(y)$  where  $m$  can be interpreted as the amount of money to be added to  $y$  to establish an indifference with  $x$ .

- ▶ If a preference relation  $\succsim$  is monotonic, continuous and quasi-linear in commodity 1, then it can be represented by a utility function of the form  $x_1 + v(x_2, \dots, x_K)$ :
  - ▶ Remember that for every  $(x_2, \dots, x_K)$  there is a number  $v(x_2, \dots, x_K)$  such that  $(0, x_2, \dots, x_K) \sim (v(x_2, \dots, x_K), 0, \dots, 0)$ .
  - ▶ By the quasi-linearity in commodity 1,  $(x_1, x_2, \dots, x_K) \sim (x_1 + v(x_2, \dots, x_K), 0, \dots, 0)$ .
  - ▶ Thus by desirability of commodity 1, the function  $x_1 + v(x_2, \dots, x_K)$  represents  $\succsim$ .
- ▶ Thus, any continuous and monotonic preference relation that is quasi-linear in the first commodity is consistent with the following procedure: the consumer considers the value (in terms of the first commodity) of a combination of goods 2, ...,  $K$ , which is independent of the quantity of the first commodity.  
 E.g. the consumer always values two apples and three potatoes at 5 dollars, independent of whether he has 100 or 1000 dollars in his pocket.